University of Camerino
MSc Computer Science (AI) · Complex Systems Design

Intelligent Search Across Organizational Knowledge

AI Puzzle piece P14 · Final presentation

QUESTION
“Can I work from home?”

FOUND, DIFFERENT WORDS
Internal smart-working regulation, Article 3
source · page · passage id

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Keyword search misses what people mean

User asks
“Can I work from home?”

Document says
“smart working arrangements”

User asks
“Fuel refund for a trip”

Document says
“mileage allowance for missions”

No shared words, so a keyword search finds nothing. But for codes and article numbers ("Art. 12, comma 3") keyword search is still the best tool.

0.16

nDCG@10 of keyword search (BM25) on paraphrased questions, out of 1.0

Measured in this project, 24 test queries

What P14 does, in four steps



1 · Ingest

Read PDF, Word, TXT and HTML safely; split into passages with stable ids



2 · Index

Turn every passage into a vector of 384 numbers (Qdrant) and a keyword index (SQLite)



3 · Search

Search by keyword and by meaning at the same time, then merge the two rankings



4 · Cite

Return passages with document, article, page and offsets



Respects permissions
Each user sees only the documents allowed for their role.



Honest when unsure
If the evidence is weak, it says so instead of guessing.

P14 is a hub of the AI Puzzle

Send documents to index

- **P11** Document reading
- **P15** Meeting memory
- **P25** Expert knowledge



P14

Intelligent search

one REST / OpenAPI contract

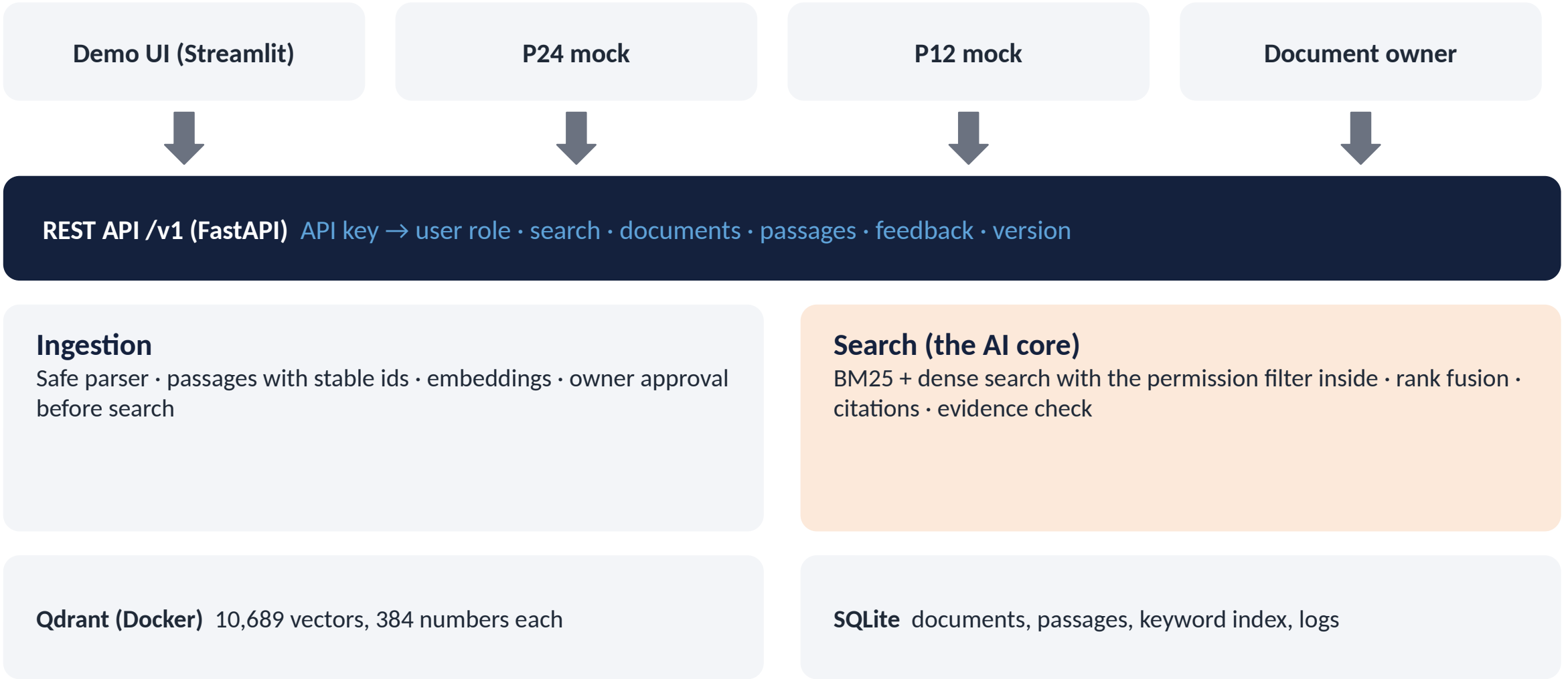


Use the search

- **P24** External assistant
- **P12** Document generation
- **P15 · P25** Meetings, experts

Useful on its own, and ready to connect. In this project P24 and P12 are implemented as working mock consumers; P11, P15 and P25 are documented.

Simple architecture: one service, two stores



Case study: a fictional public body, “Ente Demo”

32

real Italian official acts

Normattiva and Gazzetta Ufficiale; not covered by copyright (art. 5, L. 633/1941)

25

synthetic internal documents

Written by an LLM and clearly labelled; public, internal and restricted levels

10,689

passages indexed

from 57 documents in PDF, Word, TXT and HTML

Plus a public benchmark (BEIR SciFact, CC BY 4.0) used only as a sanity check. Internal documents are only 0.5% of the text: like a needle in a haystack of legislation, which makes the permission tests meaningful.

How I measured it: tune, freeze, then test

1 · Tune
40 queries
Choose the AI model and the “not sure” threshold

2 · Freeze
SHA-256 hash
Lock the 70 test queries before any tuning

3 · Test
one run
Compare three systems, report every number

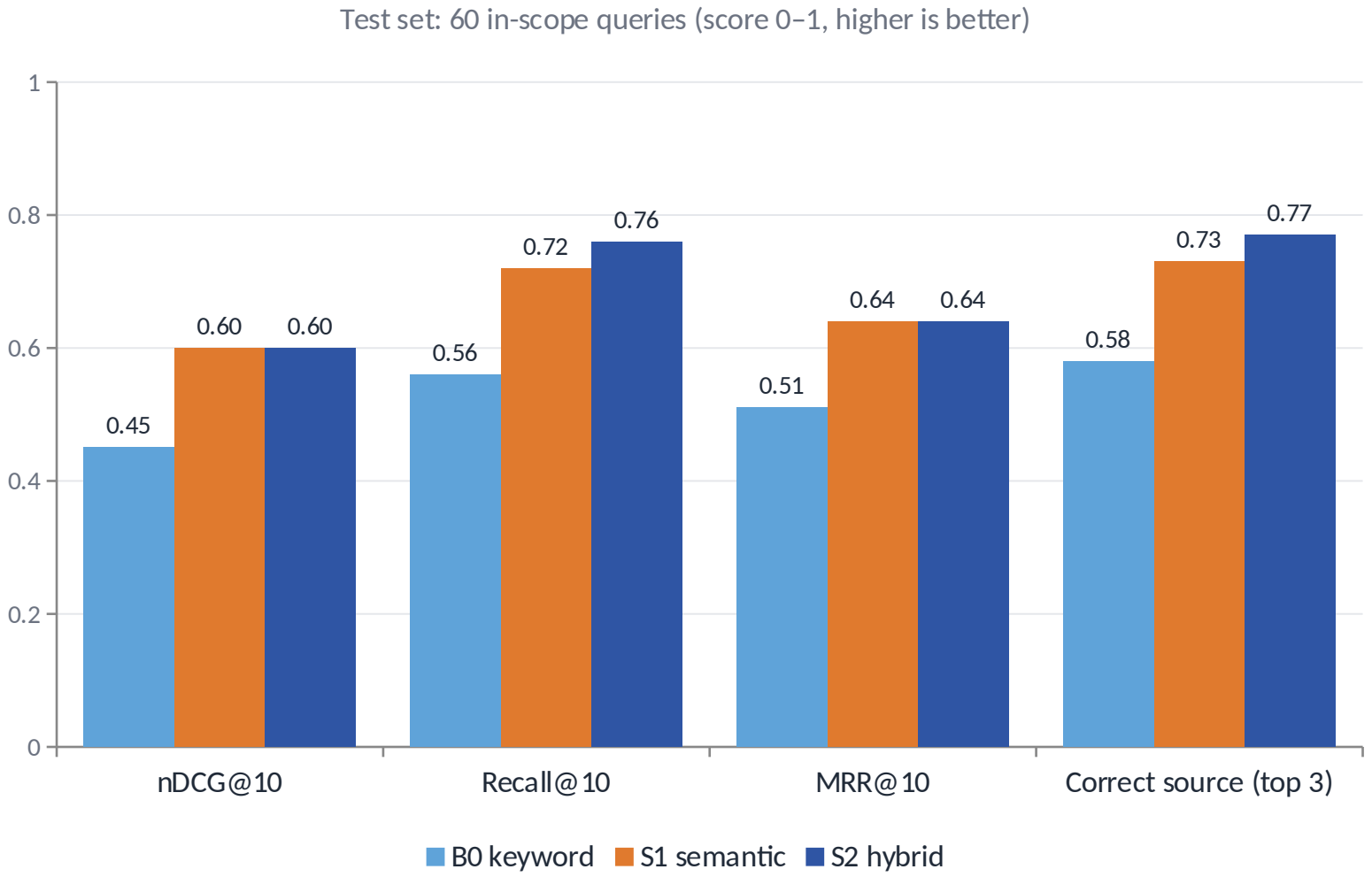
B0 Keyword search only (BM25): the baseline

S1 Semantic search only (dense vectors)

S2 Hybrid: B0 + S1 merged with rank fusion

Metrics: nDCG@10, Recall@10, MRR, correct source in top 3, latency, permission leaks. The 110 questions and their answers were drafted with an LLM: their review is my main open task.

Semantic and hybrid search clearly beat keyword search

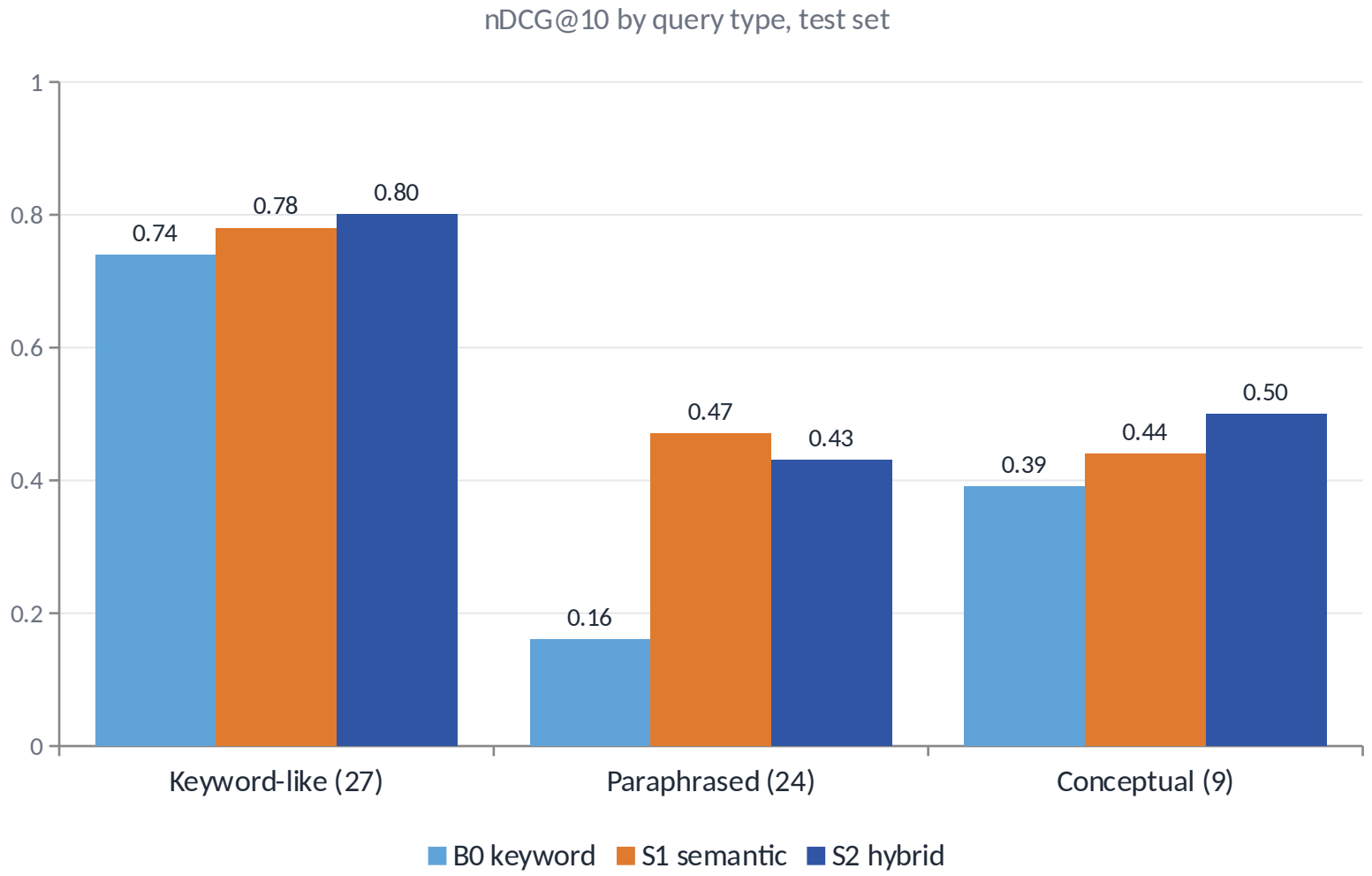


+0.15

nDCG@10 gain of hybrid over keyword search (p = 0.0001)

S1 and S2 are close on this corpus: the difference between them is not significant.

The biggest win: questions in everyday words



×3

On paraphrased questions semantic search scores 0.47 vs 0.16 for keywords.

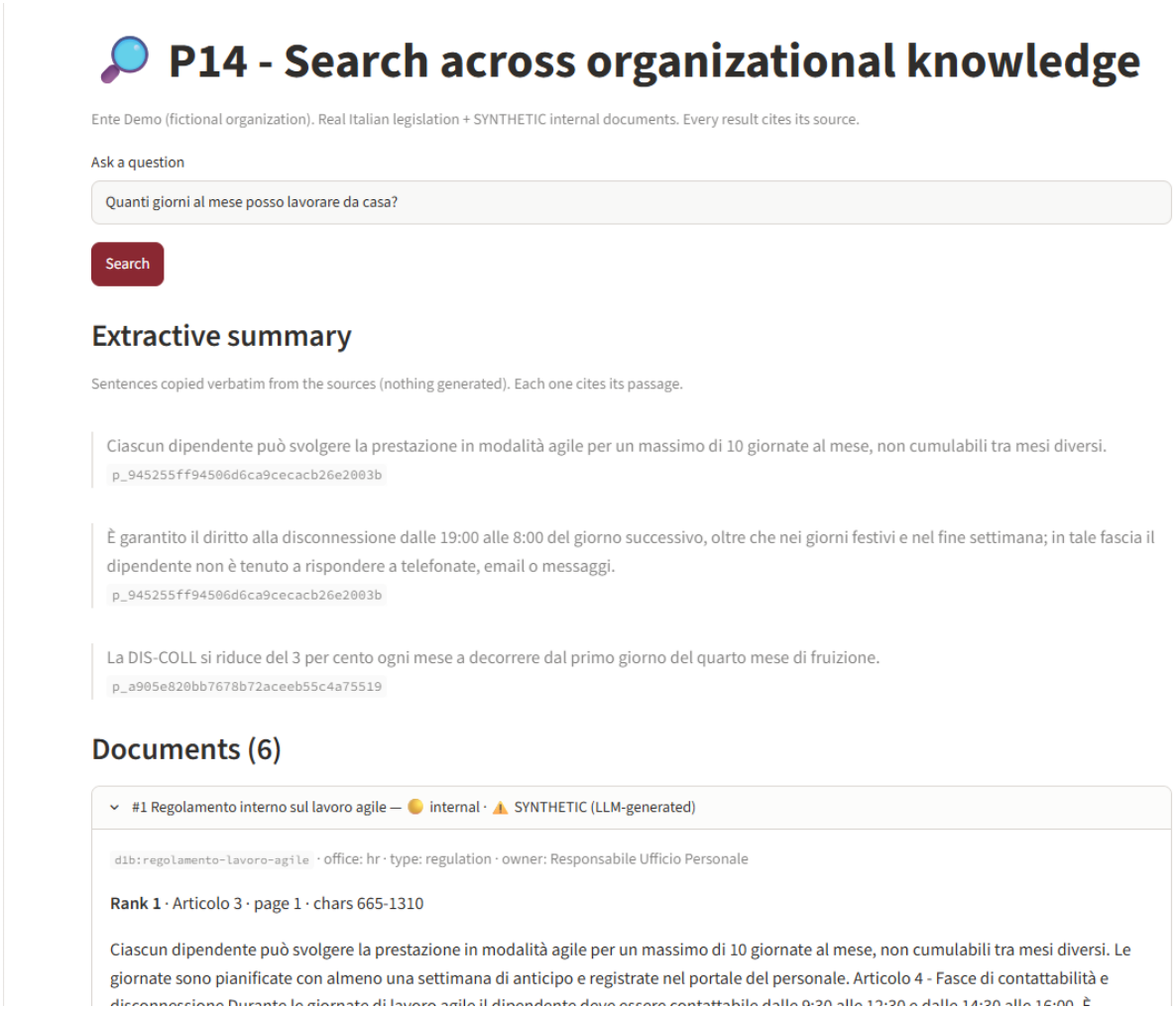
English questions over Italian documents: keyword search finds nothing, the multilingual model works.

My targets: four met, one missed

Target (mine, not official)	Measured (S2)	Result
T1 · Hybrid ≥ keyword on nDCG@10 and Recall@10	0.60 / 0.76 vs 0.45 / 0.56	Met
T2 · Correct source in top 3 ≥ 80%	76.7%	**Missed**
T3 · Correct “not sure” on out-of-scope ≥ 70%	70%	Met
T4 · Response time p95 < 1.5 s	77 ms	Met
T5 · Zero permission leaks	0 leaks in 18,576 checks	Met

Why T2 was missed: legal vocabulary that users never use, the right document but a neighbouring section, and some correct answers that were not labelled. A reranker is the next step.

Demo: every answer shows its source



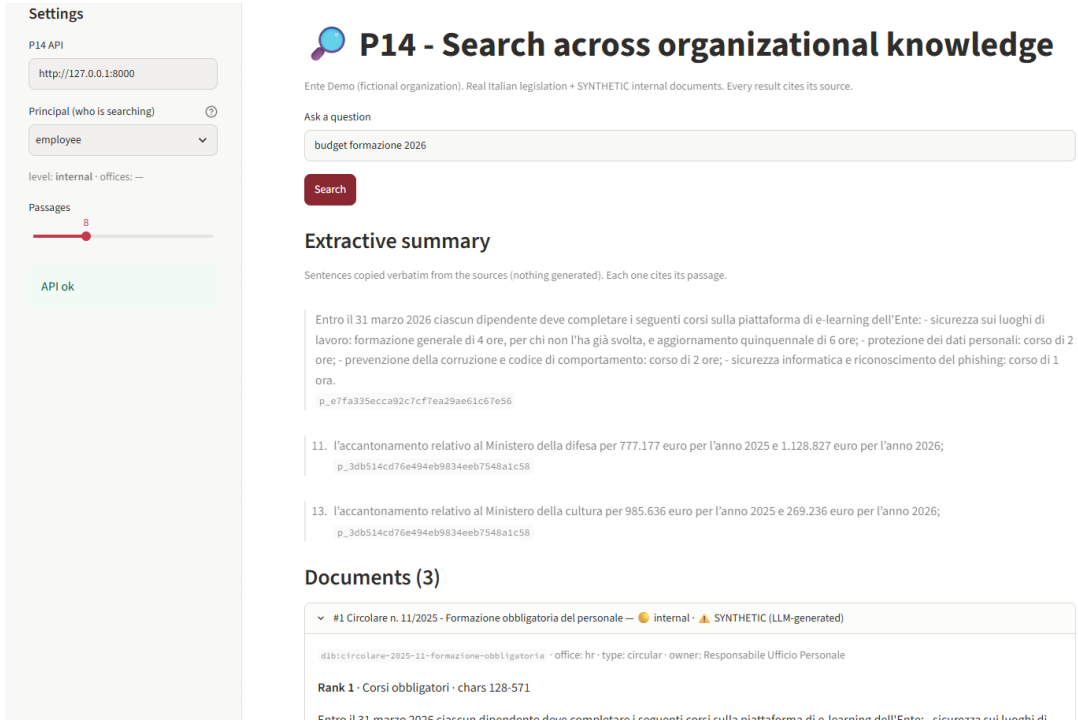
Question in everyday Italian
“Quanti giorni al mese posso lavorare da casa?”

Answer from the internal rule
Smart-working regulation, Article 3: max 10 days per month

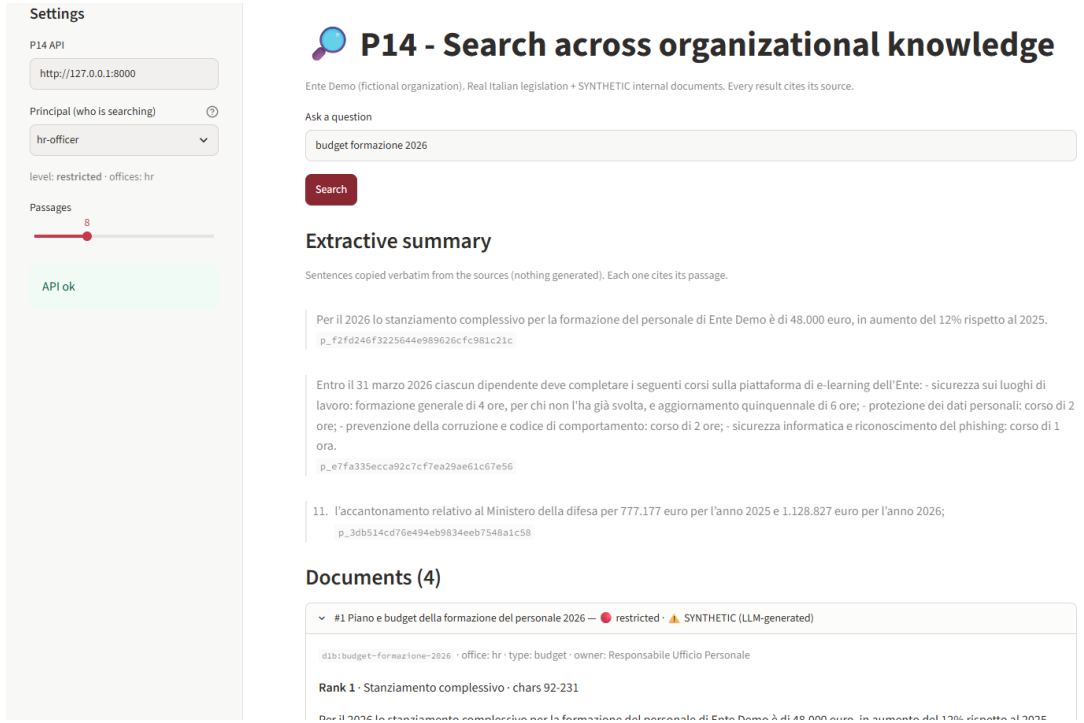
Verifiable
Every sentence carries its passage id, page and offsets

Same question, different permissions

Employee: the restricted budget is not visible

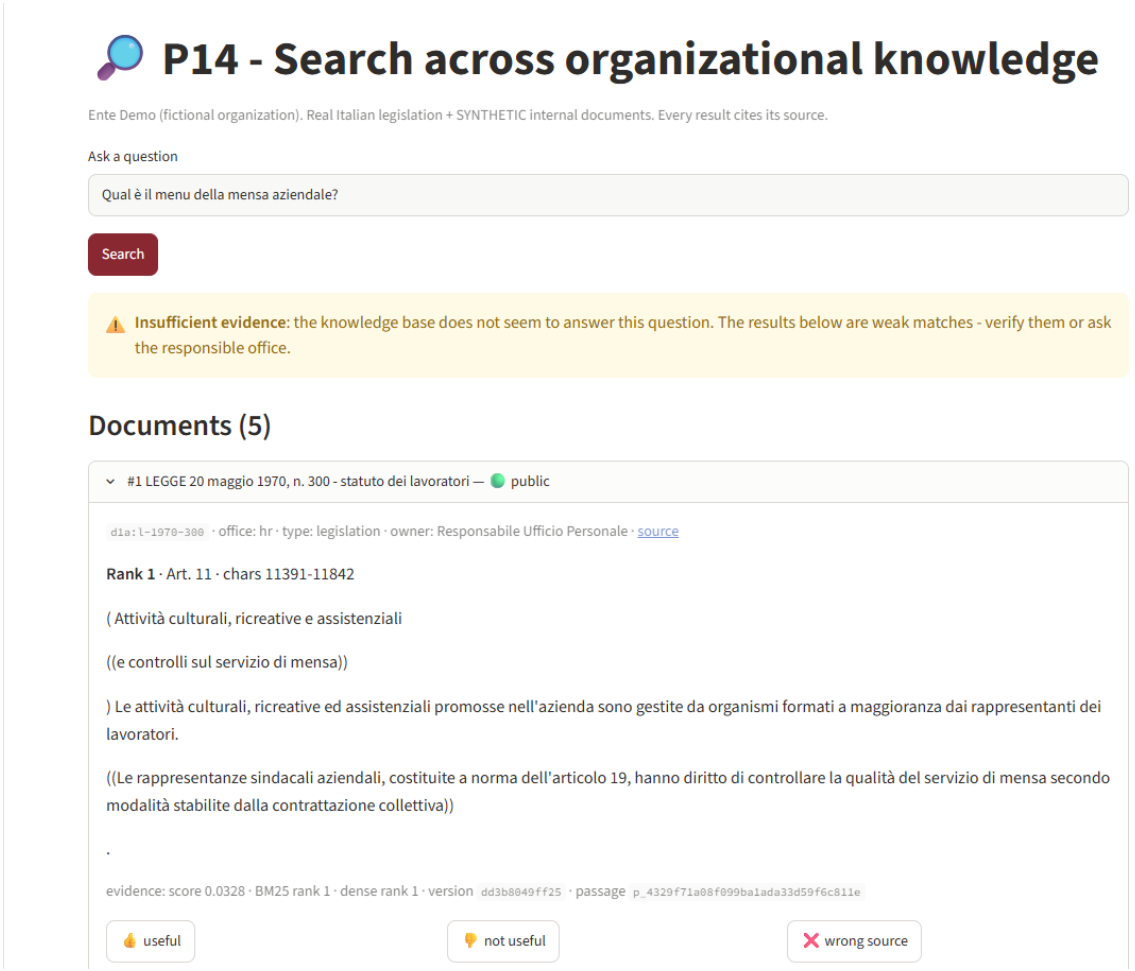


HR officer: the restricted budget is ranked first



Tested automatically: 0 leaks in 18,576 checks (9 roles × 70 queries × 3 systems).

A person stays in control



Owner approval
A document becomes searchable only after its owner approves it

Says when unsure
The canteen menu is not in the documents: the system warns instead of guessing

Everything traced
Each answer records model, index version, passages and user feedback

Shadow mode
The new system can run silently next to the old one before it is trusted

Ready to connect: a tested contract

`POST /v1/search`	ranked passages, citations, evidence, versions
`GET /v1/passages/{id}`	resolve a cited passage again
`POST /v1/documents`	ingest a document, then owner approval
`POST /v1/feedback`	store the user's judgement

OpenAPI + JSON Schemas, with real input and output examples. A test fails if the API drifts from the contract.

P24 External assistant
Sees only public passages; answers only from them; hands the case to a person when evidence is weak.

P12 Document generation
Builds a draft where every statement carries a passage id, checked again before use.

Limits and next steps

Honest limits

- No real organization: internal documents are synthetic
- Test questions and answers drafted with an LLM: need human review
- Small test set: 60 in-scope and 10 out-of-scope queries
- Real time saved by users not measured

Next steps

- Add a cross-encoder reranker to fix wrong sections
- Calibrate the “not sure” threshold per language
- Independent human judges for the test set
- A real shadow phase with real users

In one sentence

A small, fully local search system that understands meaning, always shows its sources, respects permissions, and is measured against keyword search, ready to plug into the AI Puzzle.

0.60 vs 0.45

nDCG@10, hybrid vs keyword

0.47 vs 0.16

on paraphrased questions

0 leaks

in 18,576 permission checks

47 ms

median response time

Thank you. Questions?

AI tools assisted drafting and coding; all decisions and results were reviewed by the author.